

2.5 Integration using Tables of Integrals & Computer Algebra Systems

Integration using Tables

The front and back cover of the textbook have tables of integrals. These tables show antiderivatives for more complicated integrals and can be useful at times.

Example 1: Entry number 89 in the table of integrals in your book is:

$$\int \frac{u^2}{\sqrt{a^2 - u^2}} du = -\frac{u}{2} \sqrt{a^2 - u^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{u}{a}\right) + C. \text{ Use this formula to find } \int \frac{x^2}{\sqrt{5 - 4x^2}} dx.$$

With a little manipulation we see: $\int \frac{x^2}{\sqrt{5 - 4x^2}} dx = \frac{1}{4} \int \frac{4x^2}{\sqrt{5 - 4x^2}} dx = \frac{1}{4} \int \frac{(2x)^2}{\sqrt{(\sqrt{5})^2 - (2x)^2}} dx.$

Now, we let $u = 2x \Rightarrow du = 2dx \Leftrightarrow \frac{1}{2} du = dx$. Now, we have $\frac{1}{4} \int \frac{(2x)^2}{\sqrt{(\sqrt{5})^2 - (2x)^2}} dx = \frac{1}{8} \int \frac{u^2}{\sqrt{(\sqrt{5})^2 - u^2}} du$

The formula (with $a = \sqrt{5}$) tells us

$$\frac{1}{8} \int \frac{u^2}{\sqrt{(\sqrt{5})^2 - u^2}} du = \frac{1}{8} \left(-\frac{u}{2} \sqrt{5 - u^2} + \frac{5}{2} \sin^{-1}\left(\frac{u}{\sqrt{5}}\right) \right) + C = -\frac{x}{8} \sqrt{5 - 4x^2} + \frac{5}{16} \sin^{-1}\left(\frac{2x}{\sqrt{5}}\right) + C.$$

Example 2 (you try): The following are entries which appear in the integral table in your book:

$$108. \int \frac{du}{u\sqrt{a+bu}} = \frac{1}{\sqrt{a}} \ln \left| \frac{\sqrt{a+bu} - \sqrt{a}}{\sqrt{a+bu} + \sqrt{a}} \right| + C \text{ if } a > 0$$

$$106. \int \frac{u du}{\sqrt{a+bu}} = \frac{2}{3b^2} (bu - 2a)\sqrt{a+bu} + C$$

Use one of these to find: $\int \frac{\sin(2\theta)}{\sqrt{5 - \sin(\theta)}} d\theta.$

Using a trigonometric identity we notice that $\int \frac{\sin(2\theta)}{\sqrt{5 - \sin(\theta)}} d\theta = 2 \int \frac{\cos(\theta)\sin(\theta)}{\sqrt{5 - \sin(\theta)}} d\theta$. Now, we let

$u = \sin(\theta) \Rightarrow du = \cos(\theta) d\theta$. If we use this substitution and formula 105 we see:

$$2 \int \frac{\cos(\theta)\sin(\theta)}{\sqrt{5 - \sin(\theta)}} d\theta = 2 \int \frac{u}{\sqrt{5 - u}} du = \frac{4}{3} (-u - 10)\sqrt{5 - u} + C = \frac{-4}{3} (\sin(\theta) + 10)\sqrt{5 - \sin(\theta)} + C.$$

NOTE: For class work, the table of integrals in the textbook should only be used when asked.

Computer Algebra Systems (CAS)

Computer algebra systems can integrate a wide variety of integrals. Computer algebra systems are becoming more predominant especially with free computer algebra systems like Wolfram Alpha becoming available. In this class, since Wolfram Alpha is readily available, this is what we will use.

Example 3: Use a CAS to find $\int \frac{4^x + 10^x}{2^x} dx$.

In Wolfram Alpha we type: integrate $(4^x + 10^x)/(2^x)$. The answer that it yields is:

$\frac{2^x}{\log(2)} + \frac{5^x}{\log(5)} + \text{constant}$. It makes a special note that $\log(x)$ is the natural logarithm.

Note: Sometimes the solution that we obtain from a CAS might look different than a solution that we found by hand.

Example 4: Use a CAS to find $\int \frac{\sin(2\theta)}{\sqrt{5 - \sin(\theta)}} d\theta$.

We will change the variable name to x . In Wolfram Alpha we type: integrate $\sin(2x)/((5 - \sin(x))^{.5})$. The answer that it yields is: $\frac{1}{3}(-4\sqrt{5 - \sin(x)} \sin(x) - 40\sqrt{5 - \sin(x)} + 40\sqrt{5}) + \text{constant}$. We note that this looks different from the solution we obtained in the most recent graded example. However, with some manipulation (and letting $\text{constant} = C$) we see:

$\frac{1}{3}(-4\sqrt{5 - \sin(x)} \sin(x) - 40\sqrt{5 - \sin(x)} + 40\sqrt{5}) + C = \frac{-4}{3}\sqrt{5 - \sin(x)}(\sin(x) + 10) + \frac{40\sqrt{5}}{3} + C$. Now, the

constant C can absorb the constant $\frac{40\sqrt{5}}{3}$ to form a new constant C' . Thus, we have:

$\frac{-4}{3}\sqrt{5 - \sin(x)}(\sin(x) + 10) + \frac{40\sqrt{5}}{3} + C = \frac{-4}{3}(\sin(x) + 10)\sqrt{5 - \sin(x)} + C'$. So, we see that the answer that

Wolfram Alpha provides is actually equivalent to the answer we obtained by hand.

Note: Sometimes the solution we find by hand is a much more convenient form than the solution produced by a CAS.

Note: When Wolfram Alpha uses natural logs it does so from the perspective of complex analysis (this explains the naming difference). Therefore, you might see some differences between a book solution and what Wolfram Alpha produces. In particular you might find absolute value symbols omitted, and you may even see Wolfram Alpha automatically restrict the domain of the variable which you are integrating with respect to when natural logs become involved.

Question for the Class: Can you give an example of something that you could not integrate?

Possible answers include: $\int e^{-x^2} dx$, $\int \sin(x^2) dx$, or $\int \frac{\sin(x)}{x} dx$ (Note: another example is given on page 529 of the book).

Fact: It can be proven, though it is beyond the scope of our course, that certain indefinite integrals (like those above) cannot be written using only combinations of elementary functions.

Note: The exact definition of elementary function is fairly involved, but for our purposes we can think of the elementary functions as polynomials, rational functions, power functions, exponential functions, logarithmic functions, trigonometric and inverse trigonometric functions, and hyperbolic and inverse hyperbolic functions.

Interestingly if f is an elementary function, then f' is an elementary function, but $\int f(x)dx$ need not be an elementary function!

Example: Consider $\int e^{-x^2} dx$.

Clearly $g(x) = e^{-x^2}$ is continuous and by the FTC (fundamental theorem of calculus) we know $F(x) = \int_0^x e^{-t^2} dt$ has the property that $F'(x) = g(x)$.

So, $g(x)$ does have an antiderivative (namely $F(x)$ is an example). However, it is impossible to represent an antiderivative of $g(x)$ with only elementary functions (So, someone could subjectively say that $g(x)$ has no “satisfying” antiderivative).

Note: As a side note $\int e^{-x^2} dx$ is a very important integral in statistics (in particular it is used in dealings with normal distributions).